// Vultr Agent Rush · Challenge 1: Blast Radius Zero

PressBox

Chat with your WordPress site.
Nothing reaches production untested.

Ngorbit Lab · github.com/erzawansyah/pressbox

Changing a live WordPress site is the risky part

43%

of all websites run on WordPress, mostly small businesses and the agencies behind them (W3Techs).

91%

of new WordPress vulnerabilities in 2025 were in plugins (Patchstack).

An AI agent that edits production directly

writes and runs PHP on the server that holds customer data. One mistake takes the site down.

Chat in, tested change out

01 · Ask

“Add a table reservation form to the Contact page.”

02 · Clone and contain

Fresh container, secrets redacted, no network, no keys, hard limits.

03 · Build and self-correct

Agent on Vultr Serverless Inference edits the site’s one plugin and fixes its own PHP errors.

04 · Gate

Deterministic checks decide, not the model.

05 · Preview and approve

Click the sandbox site, read the exact diff.

06 · Replay and destroy

Backup, replay, health check, auto-rollback. Sandbox destroyed.

The demo: Senja Coffee

Reservation form

The agent writes the feature, hits a PHP error, reads the log, fixes it. Gate passes, owner applies, bookings save on production.

`PASSED · 34–47 s · ≈$0.02–0.07`

Exchange rate from an external API

The agent builds the widget, but the call never leaves the sandbox. The gate rejects the unapproved domain, the sandbox is destroyed, production is untouched.

`BLOCKED · containment moment`

Live: pressbox.149-28-138-40.sslip.io · site: senja.149-28-138-40.sslip.io · judge code in the submission

Brain in the control plane, hands in the sandbox

Control plane

Agent loop, clone + redact, gate, replay with rollback. Holds the API key, has internet, **never runs agent code**.

Disposable sandbox

Redacted WordPress copy. Files + WP-CLI only. Internal network, no keys, caps dropped, limits. Destroyed per task.

Vultr Serverless Inference

DeepSeek V4.1 Flash, OpenAI-compatible tool calling.

Production WordPress + PressBox MU-plugin

Reports the web server's real PHP profile (the constraint set) and locks plugin installs in wp-admin. One theme, one custom plugin.

✕ Sandbox → internet, cloud metadata, production

→ Control plane → Vultr Inference (key stays here)

→ Production changes only by replay after approval

Blast radius: zero, by four rules

1 · Process isolation

One container per task, all Linux capabilities dropped, no-new-privileges, non-root, internal network only, no published ports.

2 · No credentials inside

Keys stay in the control plane. Passwords, sessions and secret-looking settings are redacted before the copy enters.

3 · Time and memory limits

512 MB, 1 CPU, 200 processes; PHP limits copied from production; command, step and task timeouts.

4 · Destroyed every task

Every exit path ends in docker rm -f, verified and logged. Orphans are removed on restart.

The model never approves its own work

- PHP lint passes for every file
- Every page renders with HTTP 200
- No fatal errors on re-render
- No eval, system calls, raw sockets, obfuscation, raw SQL, user changes
- No unapproved outbound domains or attempts
- PHP extensions match production
- No redacted secret in outbound traffic
- Admin accounts unchanged, no OOM kill

Judged on the **final** state: errors the agent already fixed do not count. Only then does Apply to production appear.

Vultr is the control layer

Cloud Compute

Control plane, every sandbox and the production site run on a vc2-1c-2gb VM in Singapore.

Serverless Inference

Every agent call: OpenAI-compatible chat completions with tools on DeepSeek V4.1 Flash. No other model provider at runtime.

Vultr API

The VM is provisioned by script: SSH key, instance, cloud-init. Reproducible from the repo.

The account key is used only by the provisioning script; the control plane never receives it, and the inference key never enters a sandbox.

Measured on the Vultr VM

Request	Result	Time	Cost
Reservation form on Contact	Passed, applied	32–47 s	≈ $0.02–0.07
Exchange rate via external API	Blocked by the gate	23 s	≈ $0.01
Promo loop that never ends	Blocked at 20 s limit	76 s	scripted
Change the tagline	Passed, applied	9 s	scripted

9 unit tests on the containment policies and 4 end-to-end scenarios pass; zero sandboxes left behind after any run.

Cost at Vultr’s list price for DeepSeek V4 Flash: $0.30 in / $1.00 out per million tokens.

Managed WordPress with the agent built in

Site · $19 / month

One site on its own Vultr VM, 30 agent changes. No plugins to manage, no staging to maintain.

Agency · $149 / month

10 sites, 300 changes: every change tested, logged and reversible.

~66% gross margin

$5 Vultr VM + 30 changes × ~$0.05 inference ≈ $6.50 cost per site, at Vultr list prices.

Market: the ~43% of the web on WordPress, starting with small-business sites that cannot afford a developer for every change. Prices are a hypothesis to validate.

Why not just…

	AI site builders	WordPress Studio AI	Staging site	PressBox
Changes the live site after launch	No, one-shot	Local sites only	Manual	Yes, by chat
Agent code isolated, no secrets	n/a	No	Shares server	Yes
Deterministic gate	No	No	Manual QA	Yes
Replayable change, rollback	No	No	Manual	Yes
No new third-party plugins	No	No	No	Yes

Closest hackathon entries fix generic websites or npm upgrades in sandboxes. PressBox is the one built for the 43% of the web on WordPress.

Next on Vultr

A VM per customer site, created from the app through the Vultr API.

gVisor, then microVMs on Vultr VX1 with nested virtualization.

Connect existing sites through the PressBox MU-plugin.

Parallel branches, visual diffs, MySQL support.

PressBox. Blast radius: zero.